Unit 1.01: Vectors & Embeddings in AI

Q1. Compute the dot product of the embedding vector mathbf{u} = [2, 3]^T with itself.

[x] A: 13
[ ] B: 15
[ ] C: 11
[ ] D: 10

Solution: The dot product with itself is the squared Euclidean norm: mathbf{u} cdot mathbf{u} = 2^2 + 3^2 = 4 + 9 = 13 .

Q2. Compute the dot product of the embedding vector mathbf{u} = [3, 4]^T with itself.

[x] A: 25
[ ] B: 29
[ ] C: 23
[ ] D: 14

Solution: The dot product with itself is the squared Euclidean norm: mathbf{u} cdot mathbf{u} = 3^2 + 4^2 = 9 + 16 = 25 .

Q3. Compute the dot product of the embedding vector mathbf{u} = [4, 5]^T with itself.

[x] A: 41
[ ] B: 47
[ ] C: 39
[ ] D: 18

Solution: The dot product with itself is the squared Euclidean norm: mathbf{u} cdot mathbf{u} = 4^2 + 5^2 = 16 + 25 = 41 .

Q4. Compute the dot product of the embedding vector mathbf{u} = [5, 6]^T with itself.

[x] A: 61
[ ] B: 69
[ ] C: 59
[ ] D: 22

Solution: The dot product with itself is the squared Euclidean norm: mathbf{u} cdot mathbf{u} = 5^2 + 6^2 = 25 + 36 = 61 .

Q5. Compute the dot product of the embedding vector mathbf{u} = [6, 7]^T with itself.

[x] A: 85
[ ] B: 95
[ ] C: 83
[ ] D: 26

Solution: The dot product with itself is the squared Euclidean norm: mathbf{u} cdot mathbf{u} = 6^2 + 7^2 = 36 + 49 = 85 .

Q6. Compute the dot product of the embedding vector mathbf{u} = [7, 8]^T with itself.

[ ] A: 111
[ ] B: 125
[x] C: 113
[ ] D: 30

Solution: The dot product with itself is the squared Euclidean norm: mathbf{u} cdot mathbf{u} = 7^2 + 8^2 = 49 + 64 = 113 .

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Q7. Compute the Manhattan L_1 norm and Euclidean L_2 norm of the word vector mathbf{w} = [2, -3]^T .

[x] A: |mathbf{w}|_1 = 5, ; |mathbf{w}|_2 =

[ ] B: |mathbf{w}|_1 = -1, ; |mathbf{w}|_2

[ ] C: |mathbf{w}|_1 = 5, ; |mathbf{w}|_2 =

[ ] D: |mathbf{w}|_1 = sqrt{13}, ; |mathbf{

Solution: L_1 norm is the sum of absolute values: |2| + |-3| = 5 . L_2 norm is the square root of sum of squares: sqrt{2^2 + -3^2} = sqrt{4 + 9} = sqrt{13} .

Q8. Compute the Manhattan L_1 norm and Euclidean L_2 norm of the word vector mathbf{w} = [4, -6]^T .

[ ] A: |mathbf{w}|_1 = -2, ; |mathbf{w}|_2

[x] B: |mathbf{w}|_1 = 10, ; |mathbf{w}|_2

[ ] C: |mathbf{w}|_1 = 10, ; |mathbf{w}|_2

[ ] D: |mathbf{w}|_1 = sqrt{52}, ; |mathbf{

Solution: L_1 norm is the sum of absolute values: |4| + |-6| = 10 . L_2 norm is the square root of sum of squares: sqrt{4^2 + -6^2} = sqrt{16 + 36} = sqrt{52} .